

Tone Relative to Substance in European ESG Disclosure: A Continuous Corpus-Relative Score, 2018–2024

1 Scope and contribution

Investors, lenders and supervisors read the ESG sections of corporate reports, and a recurring concern is that positive language may run ahead of the content that supports it. This paper measures that relation directly in the text: how the tone of a report’s ESG disclosure stands relative to the substance of that disclosure. Tone is the polarity of the ESG text under the finance dictionary of Loughran and McDonald (2011); substance is the density, in the same text, of mentions of a documented ESG vocabulary (§3). We compute the measure for 343 reports of 49 listed firms from seven euro-area countries, one report per firm and year over 2018–2024 (§2), a window in which European sustainability reporting moved from largely voluntary to substantially prescribed.

The measure is a continuous score, and it is relative to the corpus. Each component is standardised over the 343 reports, so a value locates a report within this corpus and has no meaning outside it; its sign says only whether the report lies above or below the corpus mean (§3.4). The idea behind the score, positive language in excess of the substance that supports it, fixes which direction of the score is of interest. The score describes disclosure, not conduct.

Two scores. We follow the architecture of Lagasio (2024): tone relative to substance, measured as a tone component minus a substance component, each rescaled across the sample. The main score, ESGSI (1), keeps that architecture and adapts its components to our corpus and question. The second score, the Extended ESGSI $\text{ESGSI}^{\text{ext}}$ (2), is the new metric this paper proposes. It adds quantification on the side of substance and hedging on the side of tone, with weights that are a choice rather than an estimate; we report how far the score moves when they change (§5).

Adapting the architecture. Lagasio (2024) introduced the index on a 2023 cross-section of 749 standalone English-language sustainability reports. Our question is how tone and substance evolve within the same firms over seven years, and four design choices follow from it.

- *Corpus and extraction.* A balanced panel needs a document that every firm publishes every year: for all but one of our firms, the annual report or the universal registration document. These filings embed ESG disclosure in a much larger financial and legal document, so we score the extracted ESG passages rather than whole reports (§3.1).
- *Substance as density.* Substance is the number of ESG mentions per 100 words of ESG text, an amount that is comparable across reports of very different length (§3.4).
- *Finance dictionary.* Tone is measured with the Loughran–McDonald dictionary, built for the language of corporate filings.
- *A continuous score.* Components are standardised as z -scores and the score is read as a position in the corpus, with no threshold (§3.4).

Because corpus, vocabulary and sentiment dictionary all differ, our scores are not numerically comparable with those of Lagasio (2024), and no such comparison is attempted.

Roadmap. §2 describes the corpus and §3 the extraction, vocabulary, scores and inference. §4 reports the scores over time, against the confound of reporting mandates, by firm group and by pillar; §5 tests their sensitivity to the choices of the method, and §6 states the limitations.

2 Data

To follow disclosure over time we need the same firms in every year, so that a change in the scores reflects a change in reporting rather than in the sample. The corpus consists of 343 reports of 49 listed firms from seven euro-area countries, one report per firm and year from 2018 to 2024. The panel is balanced: every firm contributes seven reports and every year 49. Of the 344 files collected, one 2019 report was present twice, as a re-issue with identical text, and the copy was removed by comparing document contents. For each firm-year we take one recurring annual document: an annual report (235 reports), a universal registration document (URD, 101) or, for one firm, a standalone sustainability report (7). Table 1 gives the corpus by country and document type. The document-type mix barely moves over time (12 URDs in 2018, 14 or 15 in every later year). Every measure is computed on the ESG zones extracted from each report, not on whole documents (§3.1). Firms enter the paper only through their country, their industry and anonymous positions.

Each firm carries one supersector of the Industry Classification Benchmark (ICB), assigned at firm level and not derived from the reports. The 49 firms fall into 16 supersectors and 10 of the 11 ICB industries: Financials (11 firms), Consumer Discretionary (10), Industrials (9), Consumer Staples (5), Health Care and Technology (3 each), and Basic Materials, Energy, Telecommunications and Utilities (2 each); Real Estate is absent.

Table 1: Corpus composition by country and document type. Industries: number of ICB industries represented among the country’s firms. Every firm contributes one report in each year from 2018 to 2024.

Country	Firms	Industries	Annual report	URD	Sustainability report	Total
France	17	7	13	99	7	119
Germany	14	7	98	0	0	98
Netherlands	6	5	40	2	0	42
Italy	5	4	35	0	0	35
Spain	4	3	28	0	0	28
Finland	2	2	14	0	0	14
Belgium	1	1	7	0	0	7
Total	49	10	235	101	7	343

Country and document type. Document type is almost entirely determined by country: 99 of the 101 URDs are French and all 98 German reports are annual reports, which reflects national reporting practice rather than sampling. Within the pooled panel a country effect and a document-type effect can be told apart only through 15 documents (13 French annual reports and 2 Dutch URDs), and the seven standalone sustainability reports belong to a single French firm. We treat this as a property of the design: no document-type effect on either score is identified, and since type is constant within 43 of the 49 firms, firm fixed effects absorb nearly all of it (§3.5).

Industry. Industry cuts across countries: every industry has firms in at least two countries, and France and Germany each span seven of the ten, so industry contrasts can be drawn within a single country and document regime (§4.3). Industry cannot, however, be separated from the firm: industries hold 2 to 11 firms, industry is constant within each firm and so absorbed by firm

fixed effects, and because the URD filers are 15 French firms and one Dutch firm, an industry’s URD share is essentially its French share.

3 Method

We extract the ESG passages of each report (§3.1), count in them the mentions of a documented ESG vocabulary (§3.2) and combine tone, substance, quantification and hedging into two scores (§3.3); §3.4 explains the design and §3.5 the panel inference.

3.1 Extraction of ESG passages

The reports of §2 are annual reports and universal registration documents, in which ESG disclosure is embedded in a much larger financial and legal document. Scored whole, such a document would let financial-statement bulk and document length dominate every component. We therefore first extract its ESG passages; every later measure is computed on them alone.

Each PDF is read as text blocks in reading order, and filters keep only running prose. Page furniture is dropped: blocks entirely within the top or bottom 8 % of the page, and short margin blocks that recur, digits aside, on more than 30 % of a document’s pages (running headers and footers). Tabular blocks (a high share of digits in a few lines, or short lines of which the first lacks terminal punctuation) are dropped unless they contain a vocabulary match. Short all-capital blocks are taken as section headings, which label the paragraphs that follow instead of entering the text. Navigation text (tables of contents, GRI and ESRS correspondence tables, stray page references), recognised by dotted leaders, trailing page numbers, multi-level section numbers and disclosure codes, is dropped, because such lists of ESG section names would otherwise pass the density rule below. Paragraphs are then stripped of web addresses, rejoined across line-break hyphenation and collapsed to single spaces.

Each paragraph is matched against the vocabulary of §3.2 on its unlemmatised text, ignoring case, with spaces and hyphens interchangeable between the words of a phrase and longer entries matched before the shorter ones they contain, so that *carbon footprint* counts once. A paragraph with at least one match per 100 words is retained together with one paragraph on each side; overlapping or adjacent windows are merged into a single zone, and zones shorter than 150 characters are discarded. The zones of a report, in document order, form its extracted text. Across the 343 reports, extraction kept 66,171 zones holding 28.1 million of the 71.0 million words of the PDF text layers, 39.6 % (per report, median 38.3 %, range 7.1–75.6 %). The procedure depends only on the PDF and the vocabulary, and its recall is bounded by the vocabulary: a passage that uses none of its entries is never selected.

3.2 ESG vocabulary

The vocabulary defines what counts as an ESG mention, both for selecting passages and for measuring substance, so it is documented in full. Candidate entries came from three sources: the disclosure topics and terminology of reporting frameworks and standards (GRI, SASB, TCFD and the ESRS), including the names of the frameworks and regulatory regimes under which firms label their disclosure; the ESG vocabulary of dictionary-based studies of corporate disclosure; and a structured review of sample reports from the corpus, in which independent readers listed the ESG terms they found and independent evaluators judged the pooled list. **[Author: identify the reviewers of the candidate lists.]** A candidate was admitted if its occurrences in the extracted text, read in context, predominantly carry an ESG sense (a disclosure, target, data point or policy on an environmental, social or governance matter). Candidates from the review had in addition to be dispersed across firms, with at least 50 occurrences in the reports of at least 5 of the 49 firms. A polysemous head was admitted only inside enumerated collocations (*carbon footprint* rather than *footprint*; the ESG collocations of *social* and *water*), and a morphological family or a term

carrying digits was written as a pattern (every word beginning *sustainab-* or *recycl-*; CO₂e; Scope 1–3; ISO standard numbers). Each entry belongs to one pillar: environmental (E), social (S), governance (G), or transversal (TRANS) for umbrella terms and reporting regimes (*sustainability*, *ESG*, *CSRD*). Entries whose occurrences concentrate in one industry are designated sectoral; they are part of the vocabulary, and excluding them is a sensitivity check (§5). Table 2 gives the composition.

Terms were admitted after reviewing a sample of their occurrences in context, and the frequency-weighted precision of the resulting vocabulary is estimated from those samples, not measured directly on the final extraction, at about 82 %. The stability of the scores under perturbations of the vocabulary is reported in §5.

One vocabulary drives both extraction and scoring: all 434 entries select the passages, and the 402 terms, lemmatised, are what the substance component counts. Changing the vocabulary therefore changes the text on which every component is computed, not only the substance count. For scoring, the extracted text is lowercased, tokenised and lemmatised with a standard English language model; stopwords (the standard list and 189 corpus-specific words), punctuation, numerals and tokens shorter than three characters are removed, and the vocabulary passes through the same processing, so that each entry is matched in the representation of the text it is counted on. Eleven phrases that this processing would empty or reduce to a generic word (*well-being*, both of whose words are stopwords; *say on pay* to *pay*; *ISO 14001* to *iso*) are first replaced by single protected tokens, in text and vocabulary alike. The 402 terms yield $V = 399$ distinct lemmatised scoring entries.

3.3 Components and scores

Each score combines a measure of how positively a report speaks about ESG matters with measures of how much ESG content it provides. Let $d = 1, \dots, n$ index the $n = 343$ reports and $j = 1, \dots, V$ the scoring entries. Each report has two versions of its extracted text: the *lemmatised text* of §3.2, with L_d tokens, and the *raw text*, with case, digits and stopwords intact, with R_d tokens. Let c_{dj} be the number of occurrences of entry j in the lemmatised text of d , counted over word sequences of one to four tokens (the length of the longest entry). Entries are counted independently, so an occurrence of a multi-word entry also counts for any shorter entry it contains.

Substance. SUS is the density of ESG mentions, the number of vocabulary occurrences per 100 lemmatised tokens, with every entry weighted equally:

$$\text{SUS}_d = \text{SUS}_d^{\text{dens}} = 100 \cdot \frac{\sum_j c_{dj}}{L_d}.$$

Table 2: Composition of the ESG vocabulary. Terms and collocations are literal phrases; a collocation stands in for a polysemous head. Sectoral entries cut across the three type columns. Scoring entries are the distinct lemmatised terms (patterns act at extraction only); the last column is their share of the 1,047,951 vocabulary occurrences in the extracted text of the 343 reports.

Pillar	All entries					Scoring entries	
	Terms	Colloc.	Patterns	Total	Sectoral	Entries	Occ. (%)
E (environmental)	123	29	16	168	14	152	38.2
S (social)	105	23	11	139	14	127	17.6
G (governance)	65	27	3	95	8	90	30.0
TRANS (transversal)	27	3	2	32	0	30	14.2
Total	320	82	32	434	36	399	100.0

The superscript, used in tables and figures, marks the density specification as distinct from the alternatives examined in §5.

Tone. SEN is the Loughran–McDonald polarity (Loughran and McDonald, 2011) of the lemmatised text,

$$\text{SEN}_d = \frac{s_d^+ - s_d^-}{s_d^+ + s_d^- + \varepsilon}, \quad \varepsilon = 10^{-6},$$

where s_d^+ and s_d^- count the tokens whose stem is in the dictionary’s positive and negative lists, and ε only prevents division by zero. It ranges from -1 (only negative words) to 1 (only positive words) and depends on the balance of positive and negative words, not on how many there are.

Quantification. A report that discloses substance can also put numbers on it. QUANT counts such evidence on the raw text, because preprocessing removes numerals:

$$\text{QUANT}_d = m_d/R_d,$$

where m_d is the number of matches of four pattern families: percentages; numbers of four or more digits other than years; physical units (tonnes, CO₂e, GHG, energy, power and volume units); and the names of reporting frameworks (GRI, TCFD, SASB, CSRD, SDG, the taxonomy and others).

Hedging. Qualified language (*may, could, uncertain*) weakens whatever a report asserts. The hedging component is the share of hedging words in the raw text,

$$\text{HEDGE}_d = h_d/W_d,$$

where W_d is the number of words (letter runs, hyphenated compounds kept whole) and h_d the number of them in the Uncertainty, Weak Modal, Strong Modal and Constraining lists of the Loughran–McDonald master dictionary, 495 word forms taken as written. It is computed on the raw text because the modal verbs (*may, might, could, must, will*) are stopwords. Matching is by form, not by sense: the five most frequent forms (*risk, risks, will, may, requirements*) make up 49.4% of all matches, so HEDGE registers to a large extent how often a report names risk.

Pillars. For description only, the pillar density counts the mentions of a pillar’s entries per 100 lemmatised tokens, so that the four pillar densities sum to SUS_d ; for a set of reports, the pillar share pools the mentions of the set. Neither enters a score.

Scores. Each component x is rescaled across the corpus by a z -score, $Z(x)_d = (x_d - \bar{x})/\sigma_x$, with $\bar{x}$ the mean and σ_x the population standard deviation over all 343 reports pooled across the seven years, so that yearly means of a score are comparable. The two scores are

$$\text{ESGSI}_d = Z(\text{SEN})_d - Z(\text{SUS})_d, \tag{1}$$

$$\text{ESGSI}_d^{\text{ext}} = Z(\text{SEN})_d - Z(\text{SUS})_d - w_q Z(\text{QUANT})_d + w_h Z(\text{HEDGE})_d, \quad w_q = w_h = 0.5. \tag{2}$$

ESGSI (1) is the main score of the paper: it follows the architecture of Lagasio (2024), tone minus substance with each rescaled across the sample, with the components defined above. A higher value means more positive tone relative to the ESG content of the text. The Extended ESGSI (2) is the new metric this paper proposes. Quantified disclosure is further evidence of substance, so $Z(\text{QUANT})$ is subtracted, on the same side as $Z(\text{SUS})$; hedging qualifies whatever a report asserts, so $Z(\text{HEDGE})$ is added, on the side of tone. The weights give each added

component half the weight of an original one, so that together they weigh as much as one of them and the tone–substance contrast remains the core of the score. They are a stated choice, not an estimate, since no external criterion is available to calibrate them; §5 reports the scores over a grid of weights.

3.4 Design choices

The question the scores answer is whether a report’s ESG text sounds more positive than the amount of ESG content in it would lead one to expect, compared with the other reports of the corpus. Each component is chosen for that question.

Substance is an amount per unit of text. Counting ESG mentions per 100 words means that a long report does not score more substance merely for being long, so reports of very different length are comparable; and since the count uses no statistics of the other reports, a report’s substance does not change when the sample does. As a check of convergent validity in the sense of Campbell and Fiske (1959), substance should move with quantification, which is counted with patterns on the raw text rather than with the vocabulary. Measures that keep the amount of disclosure agree with it to a similar degree (0.63 for density, 0.64 for a TF-IDF count divided by length), while a TF-IDF specification that scales each report’s term vector to unit length, and so discards the amount, agrees much less (0.30); the ordering holds, with wider uncertainty, when the shared terms are removed from both measures. §5 reports the scores under these alternative specifications. Tone is read with the Loughran–McDonald dictionary because it was built from corporate filings and classifies words by the sense they carry in that register (Loughran and McDonald, 2011).

Tone and substance come in different units, a bounded ratio and mentions per 100 words, so each is standardised before they are combined: a z -score expresses a report’s value in standard deviations from the corpus mean, and the difference in (1) is then a difference of relative positions on a common scale. The scale has no natural zero. Zero is the corpus mean of the combination, and no external benchmark exists against which a boundary could be calibrated for these reports (§6). We therefore read each score as a relative position and apply no threshold: we compare positions within the corpus through differences of means, trends, rank correlations and the overlap of the top and bottom deciles. The Extended ESGSI adds the two features of disclosure that the tone–substance contrast leaves out, whether substance is quantified and whether claims are qualified.

3.5 Inference on the panel

A firm’s seven reports resemble each other from year to year, so treating them as 343 independent observations would overstate precision. We therefore estimate trends within firms and measure their uncertainty at the level of the firm. Write $d = (i, t)$ for the report of firm $i = 1, \dots, M$ in year t , with $M = 49$. The trend is the slope of a score on calendar year with firm fixed effects, which compares each firm only with itself; in the balanced panel it equals the average of the firms’ own slopes. Standard errors are clustered by firm, with the usual small-sample correction and a t distribution with $M - 1$ degrees of freedom: they allow any correlation among a firm’s reports, but not a shock common to all firms in one year, which seven years are too few to account for, so the slope summarises the yearly means. Because 49 firms are a modest number of clusters, we also compute p -values by a wild cluster bootstrap (Cameron et al., 2008), which rebuilds the data under a zero trend by flipping the sign of all of a firm’s residuals together (+1 or -1 with equal probability, $B = 9,999$ replications); the p -value is the share of replications whose clustered t statistic is at least as large in absolute value as the observed one.

For an industry or country, the slope is the average of its firms’ slopes. With few firms, the clustered standard error of a group slope is too narrow, so we read it through the t interval on the group’s M_k firm-level slopes ($M_k - 1$ degrees of freedom). Equality of slopes across groups is

tested with a Wald test on the clustered variance, which overstates significance when a group has a single firm. Yearly means carry 95 % intervals from a regression on year indicators, clustered by firm.

4 Results

This section asks how tone relative to substance moved over 2018–2024 and why, how much of the movement reporting mandates could account for, and where firms, industries and countries stand. ESGSI (1) and $\text{ESGSI}^{\text{ext}}$ (2) are in corpus standard deviations, with trends and intervals estimated as in §3.5. Industries are ordered by mean ESGSI throughout.

4.1 The yearly series and the trend

Both scores fall in every year of the window: mean ESGSI from 0.502 in 2018 to -0.610 in 2024, mean $\text{ESGSI}^{\text{ext}}$ from 0.696 to -0.802 (Table 3, Figure 1a). The within-firm trend is -0.181 standard deviations per year for ESGSI (95% interval $[-0.250, -0.111]$) and -0.260 for $\text{ESGSI}^{\text{ext}}$ ($[-0.340, -0.180]$), and both are significant under clustered and bootstrap inference alike ($p < 10^{-4}$). The yearly intervals nonetheless overlap widely, so the slope summarises an uneven path rather than a constant rate. The crossing of zero between 2020 and 2021 is not an event, since both scores average zero over the window by construction, and the standard errors do not cover a shock common to all firms in one year (§3.5).

What moves is substance. A lower score can mean less positive tone or more substance, and it is the second (Figure 1b): SUS^{dens} rises in every year, from 5.75 to 8.05 mentions per 100 tokens, because mentions per report grow faster (+205%) than the extracted text that divides them (+119%). Tone does not move: SEN has no detectable trend ($p = 0.581$), and its yearly means dip in 2020 and rise in 2024, against the decline; a failure to reject is not a demonstrated null. $\text{ESGSI}^{\text{ext}}$ falls faster because of quantification: QUANT rises by about half over the window, though not in every year, while HEDGE is flat, its standardised yearly means within ± 0.06 of zero. In terms of disclosure practice, the reports devote a growing part of their ESG text to identifiable ESG content, and put more numbers on it, without describing it in more positive or more qualified language: the scores fall because ESG disclosure became denser, not because its

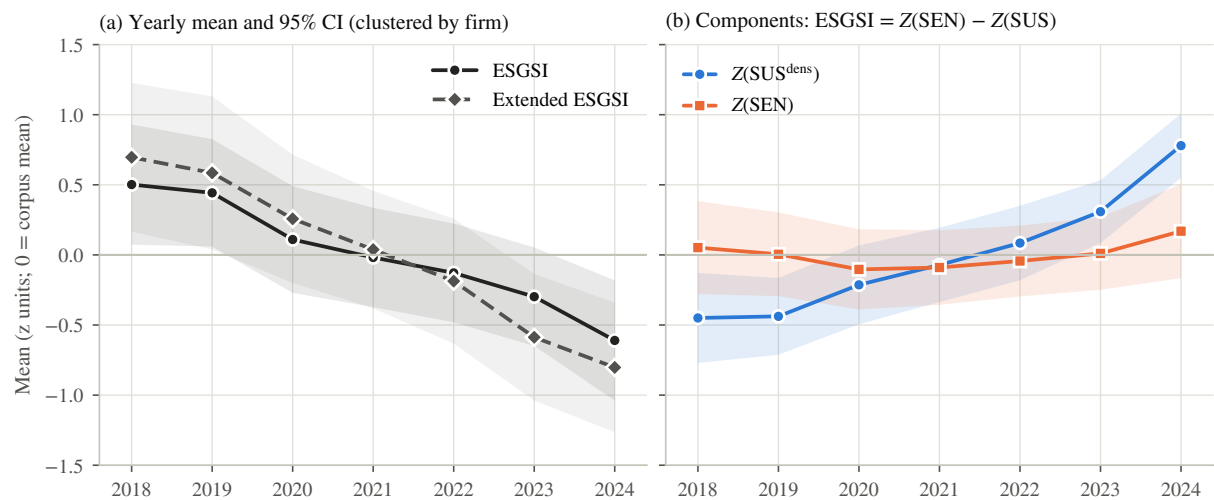


Figure 1: (a) Mean ESGSI and $\text{ESGSI}^{\text{ext}}$ by year, 49 reports per year, with 95% intervals from year indicators and standard errors clustered by firm. (b) The same for $Z(\text{SUS}^{\text{dens}})$ and $Z(\text{SEN})$, whose difference is ESGSI (1). All values in corpus standard deviations; zero is the mean over the 343 reports.

Table 3: Yearly means and trend regressions; 343 reports of 49 firms, 49 per year. Panel A: means of the two scores and of their standardised components (1), (2); z -scores are taken over the pooled corpus (§3.3), so every row averages zero over the window. Panel B: slope on calendar year with firm fixed effects; standard errors and 95% intervals clustered by firm (t_{48}); p_{wild} from a wild cluster bootstrap with 9,999 replications (§3.5); R_w^2 within firm. Component slopes are in their own units: SUS^{dens} in mentions per 100 tokens, QUANT and HEDGE in units of 10^{-4} , all per year.

<i>A. Yearly means</i>							
	2018	2019	2020	2021	2022	2023	2024
ESGSI	0.502	0.443	0.110	-0.018	-0.128	-0.298	-0.610
ESGSI ^{ext}	0.696	0.585	0.258	0.038	-0.187	-0.587	-0.802
Z(SEN)	0.052	0.005	-0.103	-0.090	-0.043	0.011	0.169
Z(SUS^{dens})	-0.450	-0.438	-0.213	-0.072	0.085	0.308	0.779
Z(QUANT)	-0.402	-0.245	-0.276	-0.073	0.123	0.522	0.351
Z(HEDGE)	-0.012	0.040	0.019	0.039	0.005	-0.057	-0.034
<i>B. Trend</i>							
	Slope	SE _{cl}	95% interval		p_{cl}	p_{wild}	R_w^2
ESGSI	-0.181	0.035	-0.250	-0.111	3.7×10^{-6}	$< 10^{-4}$	0.211
ESGSI ^{ext}	-0.260	0.040	-0.340	-0.180	3.7×10^{-8}	$< 10^{-4}$	0.295
SEN	0.0020	0.0037	-0.0053	0.0094	0.581	0.577	0.003
SUS^{dens}	0.366	0.041	0.283	0.449	1.1×10^{-11}	$< 10^{-4}$	0.364
QUANT	4.7	0.9	2.9	6.5	3.4×10^{-6}	$< 10^{-4}$	0.206
HEDGE	-0.4	0.8	-1.9	1.2	0.633	0.634	0.002

tone changed.

The panel is balanced, so the series is not a change in the set of firms or in the country mix, and the document-type mix barely moves (§2). Within Germany, fourteen firms with one country and one document type, both slopes are steeper than the pooled ones: -0.268 for ESGSI ($p = 0.002$; bootstrap 0.003) and -0.376 for ESGSI^{ext} ($p < 0.001$ under both). A trend of this sign therefore exists where country and document type do not vary, although this does not show that their mix plays no part in the pooled slope.

4.2 Regime vocabulary and the mandate confound

Over 2018–2024 European sustainability reporting moved from largely voluntary to substantially prescribed. A firm that adds a taxonomy-alignment table or a standards-referenced sustainability statement adds ESG vocabulary without necessarily changing anything it does, and SUS^{dens} cannot tell the two apart. The vocabulary names regimes that did not exist or were not in force in 2018, and the transversal pillar that holds them nearly doubles its share of all mentions, from 10% to 19% (§4.4).

The purely definitional part can be bounded by deleting entries and rescoreing, with the extraction held fixed. Deleting an explicit list of 22 regime, standard and rating names (3.8% of all mentions; deliberately over-inclusive, since several predate the window) leaves an ESGSI slope of -0.149 ($p < 0.001$), about four fifths of the original, with a rank correlation of 0.992 with the full-vocabulary score. Deleting the whole transversal pillar (30 scoring entries holding 21 of the 22 names and 14.2% of all mentions) leaves -0.117 ($p = 0.002$; bootstrap 0.003), about two thirds, with a rank correlation of 0.973. ESGSI^{ext} keeps -0.228 and -0.196 of its -0.260 . The trend is not an artefact of counting regime names.

Neither check disposes of the mandate confound. A regime elicits substantive disclosure as well as its own names (emissions, transition plans, most of the environmental vocabulary) in the words a firm that had changed its conduct would also use, and the environmental pillar is

where the rise sits (§4.4). A mandate is a common time shock, collinear with calendar year; the balanced panel, the German subsample and the fixed effects remove cross-sectional composition, not anything that moves with time. Identification would need variation in exposure, which 49 large issuers, all in scope, do not provide, so the confound is left unresolved. One consequence belongs to the architecture: since both scores fall when substance rises, a firm that adds a mandated disclosure moves down the scale for having complied. A user of the scores should therefore read the decline over this window as denser disclosure, whatever its cause, and not as evidence that firms became more candid.

4.3 Firms, industries and countries

Most of what the scores record is a difference between firms. For ESGSI, 66.7% of the variation lies between firm means and 7.2% between year means, and the correlation between two reports of the same firm, net of year (the intraclass correlation), is 0.676 (ESGSI^{ext}: 68.5%, 9.4%, 0.720). Positions persist from one year to the next and reshuffle over the window: the Spearman correlation of firms' ESGSI between consecutive years averages 0.849 (range 0.728 to 0.915) but is 0.456 between 2018 and 2024 (ESGSI^{ext}: 0.823 and 0.393). The decline is general but not uniform: 39 of the 49 firm slopes are negative for ESGSI and 43 for ESGSI^{ext}.

Industry and country are constant within firm, so every comparison between groups rests on 49 firms in cells of one to seventeen (Table 4), and every country contrast is also a document-type contrast (§2).

Levels. Industry means of ESGSI run from 1.125 (Technology, three firms) to -1.919 (Energy, two); Utilities has by far the highest SUS^{dens} and the second-lowest score. Both ends rest on two or three firms, dispersion within an industry can exceed the distance between industries (Financials' firm means span -1.41 to 2.09), and the ordering moves within the window (Figure 2): Basic Materials and Telecommunications go from well above the corpus mean in 2018 to

Table 4: Scores by ICB industry and by country, each ordered by mean ESGSI. Firms: number of firms, each contributing seven reports. Means are over the group's reports; SUS^{dens} in mentions per 100 tokens. Slope and 95% interval: group slope and t interval on the group's firm-level slopes (§3.5); none for a one-firm group.

	Firms	SUS ^{dens}	ESGSI			ESGSI ^{ext}		
			Mean	Slope	95% interval	Mean	Slope	95% interval
<i>Industry (ICB)</i>								
Technology	3	6.17	1.125	−0.328	[−0.827, 0.171]	1.268	−0.399	[−1.136, 0.338]
Telecommunications	2	5.89	0.935	−0.404	[−1.816, 1.007]	0.750	−0.495	[−1.882, 0.892]
Industrials	9	6.41	0.297	−0.137	[−0.280, 0.006]	0.200	−0.276	[−0.365, −0.188]
Consumer Discretionary	10	6.56	0.138	−0.287	[−0.471, −0.103]	0.078	−0.334	[−0.539, −0.128]
Financials	11	5.92	0.041	−0.097	[−0.221, 0.027]	0.791	−0.244	[−0.383, −0.104]
Health Care	3	6.67	−0.104	0.066	[−0.819, 0.952]	0.013	−0.019	[−0.892, 0.855]
Basic Materials	2	7.08	−0.190	−0.437	[−1.236, 0.361]	−0.731	−0.448	[−1.999, 1.103]
Consumer Staples	5	7.17	−0.575	−0.239	[−0.374, −0.104]	−0.946	−0.240	[−0.382, −0.098]
Utilities	2	9.86	−1.171	0.282	[−0.933, 1.497]	−2.573	0.499	[−3.595, 4.594]
Energy	2	7.33	−1.919	−0.286	[−0.953, 0.381]	−2.642	−0.455	[−0.716, −0.195]
<i>Country</i>								
Germany	14	5.60	1.003	−0.268	[−0.419, −0.116]	1.107	−0.376	[−0.511, −0.241]
Belgium	1	6.37	−0.203	−0.372	—	−0.043	−0.401	—
Finland	2	6.26	−0.242	−0.123	[−2.287, 2.041]	0.216	−0.399	[−0.565, −0.233]
France	17	6.84	−0.335	−0.182	[−0.296, −0.068]	−0.575	−0.250	[−0.367, −0.132]
Spain	4	8.51	−0.361	−0.055	[−0.336, 0.226]	−0.460	−0.074	[−0.361, 0.214]
Netherlands	6	7.04	−0.363	−0.181	[−0.446, 0.085]	−0.031	−0.302	[−0.570, −0.034]
Italy	5	6.65	−0.808	−0.017	[−0.360, 0.325]	−0.817	0.013	[−0.602, 0.628]

below it in 2024. $\text{ESGSI}^{\text{ext}}$ reorders the industries where quantification and hedging differ most: Financials, with the highest mean HEDGE and among the lowest QUANT, moves from 0.041 to 0.791, and Utilities, with the highest QUANT, from -1.171 to -2.573 . Within a single country and document regime, industry does not order firms reliably: across the ESGSI firm means it reaches an adjusted R^2 of 0.191 in France (17 firms, seven industries) and of -0.464 in Germany (14 firms, seven industries).

Germany stands apart among countries (1.003 against -0.335 for France), and in each of the five industries present in both countries the German mean exceeds the French one: the gap survives holding industry fixed, and so does the document-type contrast it carries. By document type, annual reports average 0.167, URDs -0.263 and the one firm’s sustainability reports -1.792 ; neither a country nor a document-type effect is identified. At report level, industry and country each add more to the explained variance of ESGSI than year does, yet firm identity still adds 0.301 of R^2 after year, industry, country and document type ($\text{ESGSI}^{\text{ext}}$: 0.259); across the 49 firm means, industry and country together reach an adjusted R^2 of 0.287 ($\text{ESGSI}^{\text{ext}}$: 0.421). Most of the difference between firms is firm-specific: a report’s position says more about its firm’s own reporting practice than about the firm’s sector or country.

Slopes. Group slopes are read through the intervals on the group’s firm-level slopes (Table 4, §3.5). Utilities shows why: its positive ESGSI slope has a clustered interval of $[0.143, 0.421]$ and a firm-slope interval of $[-0.933, 1.497]$. Read this way, the ESGSI decline is distinguishable from zero in Consumer Discretionary (10 firms), Consumer Staples (5), Germany (14) and France (17); Financials (11) and Industrials (9) have the flattest declines among the larger industries, with intervals that just include zero. Under $\text{ESGSI}^{\text{ext}}$, Financials, Industrials and the Netherlands also decline distinguishably. No group of one to three firms, including the two industries with a positive slope, supports a statement about its ESGSI slope.

4.4 Pillar composition

The rise in mentions is not spread evenly across the pillars of §3.2 (Figure 3b). The environmental share of all mentions grows from 32% in 2018 to 42% in 2024 and the transversal share from 10% to 19%, while governance falls from 39% to 23% and the social share stays between 16% and 20%. In density, environmental mentions rise in every year, from 1.73 to 3.25 per 100 tokens, and transversal ones from 0.63 to 1.52, while governance mentions fall from 2.34 to 1.90: governance loses ground absolutely, not only by dilution. The rise in substance is, in other words, environmental and regulatory (§4.2).

Across industries the composition follows the business (Figure 3a). The environmental share runs from 26% (Telecommunications) to 63% (Utilities), whose environmental density of 6.04 mentions per 100 tokens, against 4.11 in Energy, the next industry, gives it the highest SUS^{dens} in Table 4; Telecommunications and Financials are the governance-heavy industries (39% and 38%). The entries most over-represented in an industry are its own vocabulary (*hydroelectric* and *smart meter* in Utilities, *oil spill* and *methane* in Energy), so a firm scores substance partly for describing its business. The sectoral entries hold 1.0% of all mentions and at most 2.7% in any industry (Consumer Staples), too little to account for the position of Utilities or Energy. Mentions are concentrated (15 of the 399 scoring entries hold 48.1% of them); scoring without the sectoral entries, without each pillar and without the most frequent entry is examined in §5.

Reading the results. Neither score measures a quantity (§3.4): a yearly mean of -0.610 is a position within this corpus, and no level here is comparable with another application of the architecture. What is interpretable is the ordering within the sample and the direction of change: denser ESG disclosure at unchanged tone, and firms differing far more than years. None of this is evidence about conduct: the reporting mandates of the same window could move the scores with conduct held fixed, and this design cannot exclude them (§4.2).

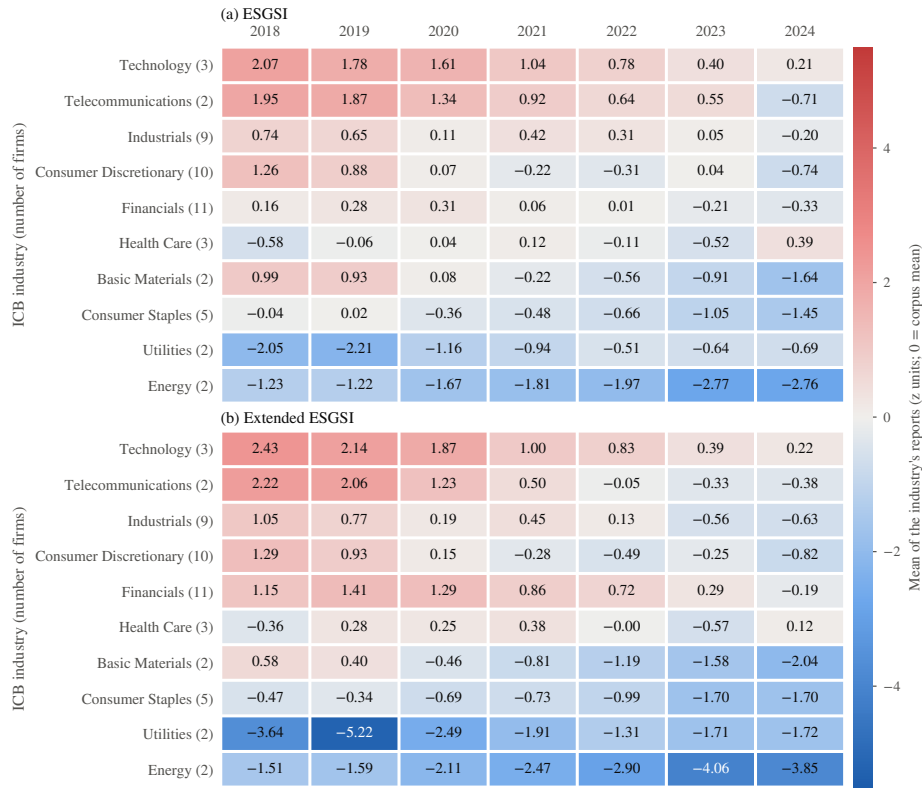


Figure 2: Mean (a) ESGSI and (b) ESGSI^{ext} by ICB industry and year, in corpus standard deviations, on a diverging colour scale centred on the corpus mean. Industries are ordered by mean ESGSI; the number of firms, in parentheses, is also the number of reports averaged in each cell.

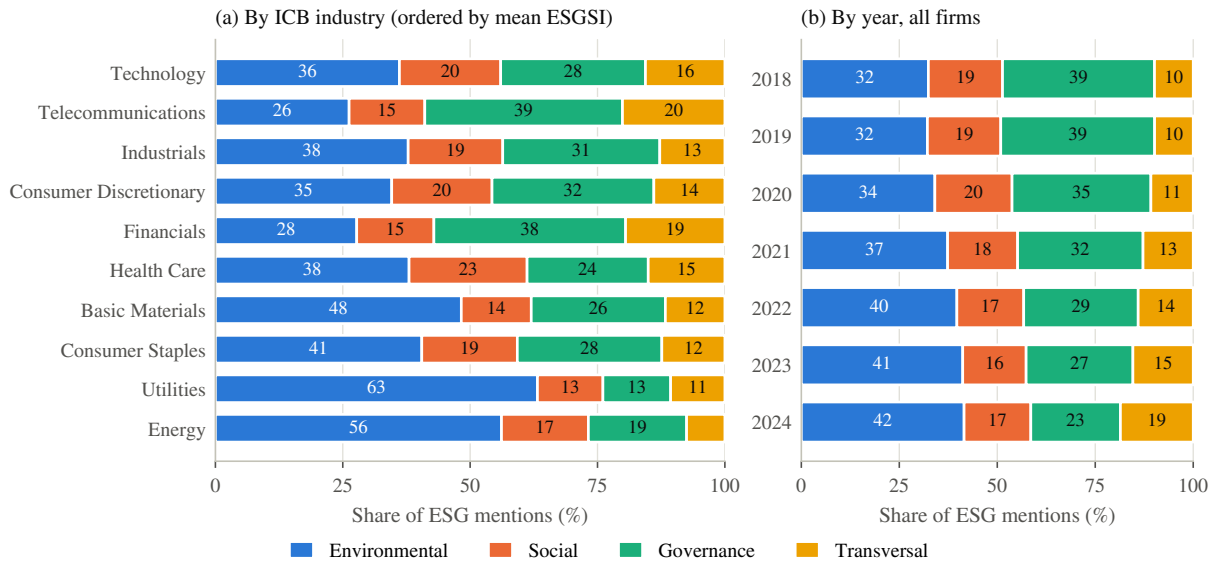


Figure 3: Share of all vocabulary mentions falling in each pillar, pooled over the group's reports, in percent: (a) by ICB industry, ordered by mean ESGSI; (b) by year, all 49 firms. The unlabelled segment (Energy, transversal) is 8%.

5 Robustness

A score built on a hand-made vocabulary and chosen weights is useful only if its conclusions do not hinge on those choices. We vary three of them—the substance specification, the weights of the Extended ESGSI and the vocabulary—and compare each variant with a reference score over the same 343 reports. Every comparison asks three questions. Is the ranking of reports preserved (Spearman rank correlation ρ with the reference)? Are the same reports at the extremes (how many of the 34 reports with the highest, or lowest, reference score remain among the 34 highest, or lowest, written top/bottom)? And does the trend survive (the within-firm slope per year, §3.5)? The reference slopes are -0.181 for ESGSI and -0.260 for ESGSI^{ext} (§4.1).

Substance specification. We recompute ESGSI with two TF-IDF versions of substance, in which each entry’s count is weighted by how few reports use it. If the weighted count is divided by the length of the text, as density is, the ranking is almost unchanged ($\rho = 0.993$; 33/32 of the extreme deciles kept): how entries are weighted matters little. If instead each report’s vector of weighted counts is scaled to unit length, the ranking changes markedly ($\rho = 0.642$; 18/21) and the slope weakens to -0.143 : how the count is scaled matters much more (§3.4).

Weights of the Extended ESGSI. Over the grid $w_q, w_h \in \{0, 0.25, 0.5, 0.75, 1\}$ of Eq. (2), ρ with the score at the default weights is at least 0.865, the minimum being at $w_q = w_h = 0$, where the score is ESGSI, and at least 0.973 within 0.25 of the default on each axis, where at least 28 of the top and 29 of the bottom 34 reports remain. The ranking depends mostly on w_h and the slope mostly on w_q (from -0.185 at $w_q = 0$ to -0.335 at $w_q = 1$, with $w_h = 0.5$); the slope is negative in all 25 pairs, and every firm-clustered 95 % interval excludes zero.

Vocabulary. The vocabulary selects the text as well as scoring it (§3.2), so each deletion of entries is run in two ways: with the extraction held fixed, so that only the list counted by SUS changes, and with a simulated re-extraction, in which the extraction rule of §3.1 is re-applied

Table 5: Deletions from the vocabulary under simulated re-extraction ($n = 343$), against the full-vocabulary score as recomputed by the simulation (slope -0.180 for ESGSI, against -0.181 for the adopted score; -0.260 for ESGSI^{ext}). Entries: of the 434. Text: share of extracted words in paragraphs that still qualify (%). Top/bot: reports of the reference top/bottom 34 that remain. Slope: within-firm slope per year. Random rows: medians over 1,000 draws (text: mean). With caveat: six frequent entries admitted although part of their occurrences is navigation text, legal boilerplate or a non-ESG sense.

Deleted	Entries	Text	ESGSI			ESGSI ^{ext}		
			ρ	Top/bot	Slope	ρ	Top/bot	Slope
Sectoral entries	36	99.6	0.9995	33/33	-0.176	0.9994	33/32	-0.256
Pillar E	168	81.1	0.8844	28/23	-0.105	0.9088	30/26	-0.211
Pillar S	139	90.1	0.9836	32/30	-0.168	0.9886	31/31	-0.253
Pillar G	95	68.6	0.9439	28/30	-0.206	0.9622	27/29	-0.250
Pillar TRANS	32	94.8	0.9822	31/32	-0.132	0.9832	32/32	-0.205
<i>Board of directors</i>	1	93.6	0.9953	33/31	-0.188	0.9965	33/33	-0.268
With caveat	6	95.2	0.9912	32/32	-0.180	0.9940	30/31	-0.255
Random 10 %	43	96.3	0.9931	33/32	-0.174	0.9954	33/32	-0.253
Random 20 %	87	91.8	0.9820	31/31	-0.165	0.9873	31/31	-0.242
Random 30 %	130	86.6	0.9674	30/30	-0.154	0.9764	30/30	-0.231
Random 40 %	174	80.8	0.9497	29/28	-0.139	0.9611	29/29	-0.214
Random 50 %	217	74.0	0.9242	27/26	-0.118	0.9396	28/28	-0.192

without the deleted entries and all components are recomputed on the paragraphs that still qualify; with the full vocabulary the simulation reproduces the adopted scores ($\rho > 0.9999$). We delete designed sets of entries and, 1,000 times at each size, random sets of 10–50 % of the 434 entries, and read each designed deletion against 1,000 random deletions of the same size. Table 5 gives the re-extraction results for both scores, and Figure 4 those for ESGSI in both modes.

Removing the 36 sectoral entries barely moves either score ($\rho \geq 0.998$ in both modes): they carry about 1 % of the vocabulary’s matches. The environmental pillar moves the score most. Without it $\rho = 0.884$, only 23 of the bottom 34 reports remain, and the slope falls to -0.105 (SE 0.039, $p = 0.009$; bootstrap 0.011), still clearly negative. Its ρ is lower than in 998 of the 1,000 random deletions of the same size, although its entries carry no more of the matches than random sets of that size (38.2 % against 38.0 %): the environmental entries order the reports differently from the rest of the vocabulary. It is the only designed deletion whose ρ falls below the median for a random half of the vocabulary (Figure 4a). With the extraction held fixed its effect is larger ($\rho = 0.776$, 12 of the bottom 34 kept, slope -0.080 with $p = 0.027$), because the paragraphs extracted for environmental entries stay in the text, unscored, and dilute density. Without the transversal pillar the slope is -0.132 ; the remaining designed deletions leave it between -0.168 and -0.206 , the steepest being without pillar G, which also removes nearly a third of the extracted text. ESGSI^{ext} moves less than ESGSI, in ρ , under every pillar deletion.

Under random deletion the ranking degrades smoothly: with half of the entries removed the median ρ is 0.924 for ESGSI and 0.940 for ESGSI^{ext}, and 27 of the top and 26 of the bottom 34 reports remain. The slope is the least stable measure. Under re-extraction its median drifts towards zero as more is deleted, from -0.174 at 10 % to -0.118 at 50 %, because less of the original text remains; with the extraction held fixed it stays between -0.179 and -0.160 . Its sign does not change: over the 5,000 random draws the ESGSI slope is negative in all but one (a 50 % deletion under re-extraction), and the ESGSI^{ext} slope in all, in both modes.

Except under unit-length scaling of substance, the ranking of reports is largely preserved and the trend stays negative; its size depends on the environmental and transversal vocabulary. These checks bound how far the scores depend on the choices varied; they do not validate the scores (§6).

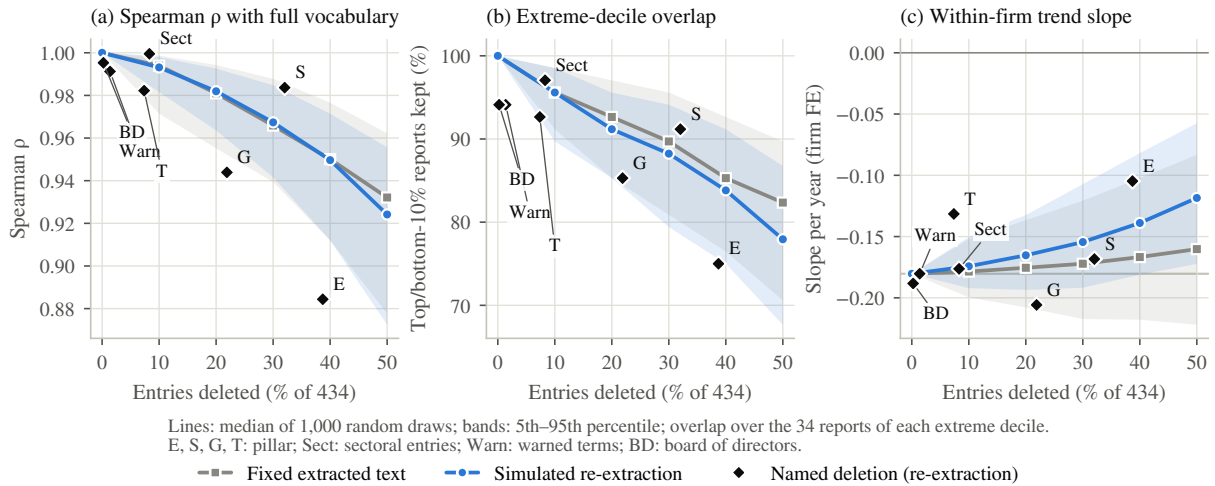


Figure 4: Stability of ESGSI under deletion of vocabulary entries ($n = 343$): random deletions in both modes, and the designed deletions of Table 5 under re-extraction, placed at the share of the 434 entries they remove (Sect: sectoral entries; T: pillar TRANS; Warn: entries with a caveat; BD: *board of directors*). (a) Spearman ρ with the full-vocabulary score. (b) Mean of the top- and bottom-decile overlaps. (c) Within-firm slope per year; the horizontal line is the full-vocabulary slope.

6 Limitations

- *No external criterion.* Both scores are positions within this corpus and would move if the panel changed. Neither is tested against an external criterion (regulatory findings, controversy data, hand-coded cases), so no value is evidence about a firm’s conduct; the one empirical check on substance, convergent validity with QUANT (§3.4), is internal and reads the same extracted text.
- *A vocabulary built on this corpus.* Candidates, admission decisions, pillars and sectoral entries were settled on the 343 reports that are scored, with no part held out, and admission rests on the judgement of the reviewers of the candidate lists [**Author: identify the reviewers.**]. Its precision is estimated rather than measured on the final extraction (§3.2), and its recall is unknown. The stability of §5 is stability around this fit; it covers deletions of entries, not additions, and the size of the trend depends on the environmental and transversal entries.
- *Country, document type and industry.* Document type is almost determined by country (§2), so neither effect is identified on its own. Industries hold 2 to 11 firms and supersectors 1 to 7, both constant within firm, so group contrasts rest on few firms (§4.3).
- *Inference.* Standard errors are clustered on 49 firms and checked with a wild cluster bootstrap, but they do not cover shocks common to all firms in a year, and seven years are too few to cluster on (§3.5).
- *Mandate.* European sustainability reporting became progressively mandated over the window, an exposure common to all firms and collinear with calendar time, so the rise in substance cannot be split into changed conduct and compliance, and compliance itself lowers both scores (§4.2).
- *Weights.* The weights of the Extended ESGSI are a choice. Its ranking is stable near them, but the size of its trend scales with w_q (§5).
- *Dictionaries.* SEN applies a finance dictionary to non-financial narrative, without negation handling, and as a bounded ratio it ignores how many sentiment words a text contains. HEDGE matches by form (§3.3): *risk* and *risks* are 147,099 of its 498,477 matches, and at least 6,632 of the 38,628 matches of *may* are the month.

Data and code availability. The code, the derived data and a runbook that regenerates every number, table and figure are available in the project repository. The source reports are public documents published by the firms.

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